INCH–POUND

MIL–L–87177A
AMENDMENT 1
28 February 1991

MILITARY SPECIFICATION

LUBRICANTS, WATER DISPLACING, SYNTHETIC

This amendment forms a part of MIL–L–87177A dated 9 February 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute "LUBRICANTS, CORROSION PREVENTIVE COMPOUND, WATER DISPLACING, SYNTHETIC"

1.1: Delete and substitute "1.1 Scope. This specification covers a synthetic, lubricant, water displacing corrosion preventive compound which may be applied from gas pressurized containers, or by dipping or brushing."

1.2.2a: Delete.

PAGE 2

2.1.1, under FEDERAL: Delete "A–A–51126, QQ–A–250/4, QQ–B–626, QQ–C–576, QQ–M–44, MIL–A–18001."

PAGE 3

Lines 1, 2 and 3: Delete and substitute "Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg 4D 700 Robbins Avenue Philadelphia PA 19111–5094."

PAGE 4

3.4.1.1 and 3.4.1.2, lines 2, after section: Add "4."

3.4.3: Delete.

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Table I: Delete and substitute.

TABLE I. Physical and chemical properties.

Characteristics	Limits
Dryness	0.0100 gram (max)
Flash point	243°C(470°F)(min)
Dielectric breakdown	25,000 volts (min)
Lubricity	1.20 mm wear scar diameter (max)
Residue soluble trichlorotrifluoroethane	No visible residue
Oxidation stability	Less than 5 pounds per 100 hrs
Grade B corrosion	No evidence of corrosion on the resistance base metal when exposed for 168 hours in accordance with methods s specified in section 4.
Sprayability	Sprayable.

Grade C shall meet all the physical and chemical properties in table I except the corrosion resistance (Grade B) requirements.

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Table II: Delete and substitute.

TABLE II. First article testing 1/

Characteristics	Requirements	Test method	ASTM
Dryness	3.3, table I	4.6.1	
Flash point	3.3, table I		D1310
Dielectric breakdown	3.3, table I		D877
Lubricity	3.3, table I		D2266 2/
Residue soluble in trichlorotrifluoroethane	3.3, table I	4.6.3	
Leakage	3.4.1.1	4.6.4	
Content	3.4.1.2	4.6.5	
Performance of pressurized containers	3.4.1.2	4.6.6	
Oxidation stability	3.3, table I		D942
Grade B corrosion resistance	3.4.4	4.6.8	
Sprayability	3.3, table I	4.6.9	
Grade C shall meet all the first article testing in table II, except the corrosion resistance (Grade B) requirements.			

1/ Refer to 4.3.1

2/ Compound shall be weathered before loading into ball pot.

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Table III, lines 3 and 4: Delete "Grade A Corrosivity 3.4.3 4.4, 4.5, 4.6, 4.6.7".

4.6.2: Delete.

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4.6.7, 4.6.7.1, 4.6.7.1.1, 4.6.7.1.2, 4.6.7.2: Delete.

PAGE 12

Line 3, from top of page, delete and substitute: "LUBRICANTS, CORROSION PREVENTIVE COMPOUND, WATER DISPLACING, SYNTHETIC GRADE ____".

PAGE 13

6.3: Delete and substitute "Typical formulation of compound for Grade B. Grade B is intended for applications where severe corrosion is possible. Grade B can be used for control of surface static electricity and as a barrier film to minimize surface contamination".

6.3.1: Delete.

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Table IV: Delete.

Custodians:
Army – ME
Navy – AS
Air Force – 11

Preparing activity:
Air Force – 11

Review activities:
Air Force – 68
DLA – GS

User activities:
Army – AR
Navy – EC

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